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Democratization or Diversion?

The Effect of Community Colleges on Educational Attainment

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This article estimates the impact of community colleges on educational attainment. Community colleges give those who might not have otherwise attended college the ability to do so (*democratization*); they also attract students who might otherwise have attended a four-year college (*diversion*). The effect of democratization on educational attainment is nonnegative; the effect of diversion is unclear a priori. I use variation in access to community colleges as instrumental variables to address self-selection into types of colleges. The results suggest that two-year college students diverted from a four-year college complete fewer years of education. Overall, however, community colleges appear to increase total years of schooling but do not appear to change the likelihood of attaining a bachelor's degree.

KEY WORDS: College accessibility; College proximity; College tuition; Instrumental variables; Junior college; Two-year college.

1. INTRODUCTION

Junior colleges were first designed in the late 19th century as local, relatively easy-access schools giving all who wanted to attend college the opportunity to do so without overburdening the four-year colleges. Students could complete their first two years in a junior college and earn a two-year degree (an associate's degree); those who chose to do so could continue in a four-year institution and complete their last two years. (I use the terms "junior college," "two-year college," and "community college" interchangeably.) The junior-college model has survived and is thriving. During the late 1970s and early 1980s, more than 50% of all first-time, first-year college students started in a two-year school. In 1987, 35% of 18–19-year-old college students and 30% of full-time equivalent college students attended junior college (*Digest of Education Statistics*, 1989, pp. 171, 178, and 180).

In this article I address two questions that have long been asked of junior colleges: Does improving the accessibility of community colleges increase educational attainment? Does starting in a community college, rather than a four-year college, affect an individual's educational attainment? The first question is addressed using the following reduced-form model of educational attainment (the structural model will be presented later):

$$BA_i = X_i\beta + \Gamma_2 D_{2i} + \Gamma_4 D_{4i} + \varepsilon_i, \quad (1)$$

where BA_i is a measure of individual i 's educational attainment; X_i are individual-specific covariates; D_{2i} and D_{4i} measure accessibility of two-year and four-year colleges, respectively; ε_i is an error term; and β , Γ_2 , and Γ_4 are coefficients. From this equation, Γ_2 measures the overall effect of improving two-year college accessibility on educational attainment.

If educational attainment is measured as college attendance, then increased accessibility or closer proximity should decrease costs (because the individual could live at home, saving on room and board, and because the direct outlays in tuition would be lower), thereby increasing the likelihood that the individual attends college. The overall effect of junior-college accessibility on other forms of educational attainment is not so clear. To see this, consider the following structural model:

$$BA_i = X_i\beta_0 + \alpha_2 A_{2i} + \alpha_4 A_{4i} + \lambda S_i + \mu_i, \quad (2)$$

$$A_{2i} = X_i\theta_2 + \gamma_2 D_{2i} + \delta_2 D_{4i} + \lambda_2 S_i + \mu_{2i}, \quad (3)$$

and

$$A_{4i} = X_i\theta_4 + \gamma_4 D_{2i} + \delta_4 D_{4i} + \lambda_4 S_i + \mu_{4i}, \quad (4)$$

where many variables are as defined previously and A_{2i} signifies starting at a junior college, α_2 signifies the average effect of starting at a junior college on educational attainment, A_{4i} and α_4 are analogous measures for starting at a four-year college, S_i represents the individual's (unobserved) desired level of schooling, and μ_i , μ_{2i} , and μ_{4i} are error terms. The base group consists of those who have not attended any college. Solving for the reduced-form of this structural model, it is clear that the coefficient, Γ_2 , implicitly combines two effects. Specifically,

$$\Gamma_2 = \alpha_2 \gamma_2 + \alpha_4 \gamma_4. \quad (5)$$

Several factors determine the overall effect of community colleges on educational attainment. First, the coefficients, γ_2 and γ_4 , from Equations (3) and (4) reflect how accessibility changes college enrollment decisions. On the one hand, community colleges may provide a place in higher education for those who otherwise might not attend college either for

financial reasons or because they were not yet ready to attend four-year college. Brint and Karabel (1989) termed this the *democratization effect*. On the other hand, local community colleges may also draw away students who might otherwise have attended a four-year college, the *diversion effect*.

The reduced-form effect of accessibility on educational attainment will also depend on the effect of starting at each of the two schools on educational attainment—that is, α_2 and α_4 . Deriving consistent estimates of these effects will be the focus of the second part of the analysis. There are at least two reasons why policymakers might be concerned that $\alpha_2 \neq \alpha_4$ and that diversion from a four-year college affects educational attainment. First, leaders of the California state university system have considered limiting enrollments in the four-year colleges and encouraging students to begin at a two-year college (Trombley 1991). A key question is whether such a policy would affect the educational attainment of those students denied admission to a four-year college.

Second, consider S_i , an individual's desired schooling or any other unobserved variable that is correlated with the actual educational attainment as well as with the type of college selected. Assume that this unobserved characteristic is fixed over time. If educational outcomes of students who begin in a community college only differ from those who begin in a four-year college because the two-year college students desire less education, then, conditional on desired level of schooling, $\alpha_2 = \alpha_4$ and policymakers need not concern themselves with these parameters. If two-year-college students receive less schooling, conditional on desired educational attainment, such that $\alpha_2 < \alpha_4$, however, policymakers might ask why. One could argue that an individual's desired level of schooling changes over time and that this change is unrelated to the type of institution that the individual attends. Again, if this is the case, policymakers need not be concerned about differences between the two types of institutions. It is also possible, however, that the difference is due to some effect of community colleges. Clark (1960) and Brint and Karabel (1989) argued this case in claiming that the vocational-education and terminal-degree programs of community colleges are not conducive to completing four years of college, even for those who aspire to a four-year college degree. Essentially these authors argued that many students lack the necessary information to make an informed decision between two- and four-year college. Although the potential importance of college type (and hence the effects of diversion) on educational attainment is an empirical issue, econometrically it is difficult to estimate because desired levels of schooling are inherently unobservable.

The goal of this article is to estimate the overall effect of community colleges on educational attainment and to obtain a consistent estimate of the effect of college type. I use distance to two-year and four-year college and average state two-year and four-year college tuition as measures of college accessibility in a reduced-form equation and then use them to instrument for college attendance in the corresponding structural equation. If proximity and cost are not correlated with

unobserved factors that affect educational attainment, conditional on college type and other observable covariates, I am able to estimate a consistent effect of two- and four-year colleges on educational attainment. The results suggest that, contrary to earlier findings, closer two-year college proximity is associated with higher rates of college attendance. Furthermore, overall, junior colleges may marginally increase aggregate years of schooling but do not appear to change the likelihood of attaining a bachelor's degree. I estimate that the negative effect of starting at a two-year college on years of education for those students who were diverted from a four-year college is much smaller than that previously estimated by others. These conclusions must be interpreted as suggestive, however, because the measures of accessibility, though theoretically appealing, are not entirely satisfactory.

2. THE SAMPLE

The sample consists of 6,786 individuals from the High School and Beyond (HS&B) senior cohort who participated in all four rounds of the survey. (See the Appendix for a more complete description of the data.) Junior colleges typically have open admissions, requiring either a high-school diploma or an "ability to benefit" from attendance. All members of the high-school class of 1980 were thus eligible to attend. I should note, however, that many high-school dropouts attend junior colleges to receive a high-school equivalency diploma (GED). Because all participants in the survey reached the spring of their senior year in high school, this study is likely to underestimate the overall value of community colleges in increasing educational attainment.

I categorize respondents according to the first post-secondary institution attended after high school, two-year college or four-year college. I include measures of gender, race (Black, Hispanic, and Other), and socioeconomic status—parents' education, family income, and whether the family owns their own home. To provide another socioeconomic indicator, I include the percentage of the high school's enrollment that is White. Measured ability is captured by the composite score from a battery of tests administered during the spring of the senior year in high school to all respondents. Local labor-market indicators proxy for the opportunity cost of staying in college. I include the city unemployment rate and state average manufacturing wage (\$1,982) in the respondent's state in 1980. Finally, census region, whether the high school is in an urban area, and income per capita (\$1,982 in the respondent's high-school state of residence) control for location effects that may be correlated with the proposed instrumental variables.

I primarily use two measures of "educational attainment." The first is attainment of a bachelor's degree because the traditional mission of junior colleges is to allow students to transfer to four-year schools and because there are clear economic gains associated with graduating from a four-year college. Because not all students aspire to the bachelor's degree, I also use total (completed) years of college education. The means and standard deviations of the variables are in Table

Table 1. Sample Means and Standard Deviations

Variable	Started in two-year college		Started in four-year college		Control group	
	Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.
Female	.57	.49	.56	.50	.55	.50
Black	.17	.37	.22	.41	.25	.43
Hispanic	.25	.43	.14	.35	.20	.40
Test score	48.88	8.01	54.40	8.00	44.79	7.60
Father's education:						
Missing	.19	.39	.15	.36	.25	.43
Vocational school	.07	.26	.07	.25	.07	.25
Some college	.12	.32	.11	.31	.05	.23
Has bachelor's degree	.15	.35	.27	.44	.05	.22
Mother's education:						
Missing	.11	.32	.07	.25	.16	.37
Vocational school	.09	.29	.10	.29	.06	.24
Some college	.14	.34	.15	.35	.06	.23
Has bachelor's degree	.09	.29	.20	.40	.04	.19
Family income (\$80)	20,141	15,634	23,538	18,373	15,930	13,783
Family income missing	.08	.27	.08	.27	.13	.33
Own home	.79	.41	.80	.40	.73	.44
% high school enrollment that is White	64.61	32.16	68.83	33.02	66.90	34.14
Urban	.25	.44	.26	.44	.25	.43
City unemployment rate	7.42	2.65	7.52	2.68	7.75	2.78
State income per capita (\$82)	11,350	1,194	11,046	1,200	10,731	1,248
State avg. manufacturing Wage (\$82)	9.29	1.30	9.59	1.32	9.46	1.41
Northeast	.14	.35	.26	.44	.19	.39
Midwest	.19	.39	.28	.45	.26	.44
South	.35	.47	.30	.46	.39	.49
West	.33	.47	.15	.35	.15	.36
Completed bachelor's degree by 1986	.11	.32	.43	.49	.01	.08
Years of education	13.51	1.49	14.89	1.59	12.30	.87
Miles to two-year college	10.72	16.29	17.17	28.56	21.54	29.76
Miles to four-year college	18.00	23.36	14.65	20.37	21.18	25.44
Two-year tuition in state (\$82)	389	296	543	281	514	267
Four-year tuition in state (\$82)	711	356	861	321	833	322
Vocational/technical school	NA		NA		.25	.43
Number of observations	1,689		2,748		2,349	

3. THE EFFECT OF COLLEGE ACCESS ON EDUCATIONAL ATTAINMENT

To study how college accessibility affects educational attainment, I estimate Equation (1) using miles from the respondent's high school to the nearest two-year and four-year colleges as well as average two-year college tuition and average tuition at comprehensive four-year schools in the state of the respondent's high school in 1980 as measures of college accessibility. Ideally, one would study this question by randomly choosing neighborhoods in which to build community colleges or randomly assigning tuition levels. In reality, people who value education may move to areas in which there are more colleges. If residence is also correlated with unobservable variables that determine educational attainment, then estimates from this equation will likely be biased upward. Such a correlation is mitigated by the fact that both measures of accessibility are based on residence when respondents were seniors in high school so that residence and college attendance were not contemporaneously determined and primarily reflect the decisions of the parents. I further

control for this to some extent by conditioning on observable measures of family background, as well as whether the high school is in an urban area.

The results are presented in Table 2. This first two columns update earlier work by Anderson, Bowman, and Tinto (1972) on the role of college proximity in educational attainment. The farther the nearest two-year and four-year colleges, the less likely it is that an individual will attend college. Tuition, particularly two-year college tuition, appears to play a less significant role than proximity in the decision to attend any college. The *F* statistics presented suggest that, together, the measures of college accessibility are a significant determinant of college attendance. Note that the estimates in column (1) do not control for measures of family background, but those in column (2) do. The fact that the coefficients decrease in all cases is some evidence that college proximity may be correlated with educational attainment and that the case for exogeneity can only be made conditional on family background and location effects, a point to which I return in Section 5.

Table 2. The Effect of Community College Accessibility on Educational Attainment

Independent variables	Dependent variable					
	College attendance (1 = attend/0 = don't attend)		Years of education		Ever-attain degree (1 = bachelor's degree/0 = no degree)	
	(1)	(2)	(3)	(4)	(5)	(6)
Miles to two-year college (÷10)	-.014 (.003)	-.011 (.002)	-.026 (.009)	-.011 (.007)	-.004 (.002)	-.001 (.002)
Miles to four-year college (÷10)	-.016 (.003)	-.007 (.003)	-.057 (.010)	-.018 (.009)	-.009 (.002)	-.002 (.002)
Two-year college tuition (\$82) (÷1,000)	.041 (.050)	.009 (.044)	.343 (.185)	.212 (.157)	.082 (.043)	.061 (.039)
Four-year college tuition (\$82) (÷1,000)	-.060 (.038)	-.058 (.033)	-.346 (.136)	-.348 (.115)	-.021 (.031)	-.024 (.028)
Includes family background		X		X		X
F statistic of joint significance of accessibility variables	15.66	8.39	12.09	3.84	6.59	.99
R ²	.046	.239	.035	.291	.034	.189

NOTE: There are 6,786 observations. White (1980) standard errors are in parentheses. Also included are a constant, female, race, % of high-school enrollment that is White, high school in urban area, high school's city unemployment rate in 1980, census region dummies, and state income per capita in 1980 in state of residence when in high school. "Family background" includes the HS&B test score, parents' education, family income, and whether family owns home. The coefficients from the other independent variables are available from the author on request.

The equations in the final four columns estimate the effect of college accessibility on other measures of educational attainment. The farther an individual lives from both two-year and four-year colleges, the lower his or her educational attainment. Two-year college proximity, however, has a *p* value of only .107 when family background measures are included in column (4); four-year college proximity remains significant at conventional levels. Furthermore, the magnitude of both coefficients is quite small. In all cases, two-year college tuition has the "wrong" sign, suggesting that conditional on four-year college tuition, lower two-year college tuition lowers educational attainment. This effect is not significant once family background is included. In general, the coefficients on tuition are not significant conditional on family background, and these standard errors are only likely to understate the true standard errors given that state tuition is an aggregate measure (Moulton 1990). Finally, the results in column (4) suggest that college accessibility has virtually no effect on finishing a four-year degree.

In all, the results indicate that, contrary to the results reported by Anderson et al. (1972), increasing both two-year and four-year college accessibility increases the likelihood of attending college. Furthermore those who live near a community college appear to complete slightly more years of education, although college proximity appears to have only a small (positive) effect on the likelihood of attaining a bachelor's degree. These estimates represent the overall effect of community colleges on educational attainment. As shown in Equation (5), however, these coefficients do not indicate whether the type of college attended affects educational attainment.

4. IDENTIFYING THE EFFECT OF COLLEGE TYPE

The ideal manner in which to evaluate the effects of college type on educational attainment would be to conduct a random

experiment. For example, a group of high-school guidance counselors would randomly assign some fraction of those students who express a desire to attend four-year college to start in a two-year college. The control group would consist of those who were assigned to start at a four-year college. After some period of time, educational attainment would be measured and a treatment effect, the difference in mean outcomes between the control group and the treatment group, would be evaluated. Alba and Lavin (1981) attempted such an experiment in their study of applicants to four-year schools in the City University of New York system. They tracked applicants admitted to the four-year system and those assigned to a two-year school, supposedly by clerical error, and found that students assigned to a two-year college had lower educational attainment than the four-year college students. There is evidence (in the preadmission means), however, of unobserved heterogeneity in students starting at the two types of colleges.

Other authors have interpreted ordinary least squares (OLS) results as identifying the treatment effect by assuming that, conditional on the included covariates, there are no additional factors related to both attending junior college and educational attainment. (See Anderson 1981; Breneman and Nelson 1981; Velez 1985.) They generally find that junior-college students receive less education than similar four-year college students. Although conditioning on measures of family background goes part of the way toward addressing self-selection, test scores and other observable measures of ability are unlikely to fully capture one's desired level of schooling and academic ability. In addition, these studies evaluate the effect of starting at a junior college on attainment by comparing two-year to four-year college students. Although their interest is in isolating the effects of college type, the authors observe only those who did attend junior college, a group that includes some who would otherwise have attended no college at all and who may not aspire to attend a four-year college. Even if the diversion effect has no impact on educational

attainment, conditioning on college attendance can result in a lower estimate of the educational impact of junior colleges relative to four-year colleges (Rouse 1992).

I improve on this previous literature by attempting to estimate the effects of both diversion and democratization on overall educational attainment and by estimating the structural model [Eqs. (2)–(4)] using an instrumental variables (IV) strategy. The success of the IV strategy hinges on the validity of the instruments. Equations (3) and (4) represent the first-stage equations in which miles from the respondent's high school to the nearest two-year or four-year college and the average state two-year or four-year college tuition are the measures of college accessibility. Clearly, the same caveats regarding the possible endogeneity of these measures of accessibility (as discussed in Sec. 3) apply here as well; if residence is also correlated with unobservable variables that determine educational attainment, then the instruments are not valid.

Table 3 shows results from linear probability models on the first-stage specification. Conditioning on family background and state variables, those living closer to a junior college are more likely to begin in one, and the relationship is nonlinear. The effect of distance to a two-year school has an opposite and significant effect on four-year college attendance. The own-price and cross-price effects of tuition on college attendance also have the right signs and are significantly different from 0. The *F* statistics of the joint significance of the excluded variables are also reported in the table. Instruments with little explanatory power in the first stage can generate bias in the IV estimates (Bound, Jaeger, and Baker 1993). The *F* statistics of 41.16 and 14.22 suggest that bias in the IV estimates due to little explanatory power of the instruments is likely to be minimal.

Table 3. First-Stage Results (linear probability)

Independent variables	Dependent variable: two-year college four-year college	
	Coefficient (std. error)	Coefficient (std. error)
Miles to two-year college (÷10)	-.036 (.004)	.013 (.004)
Miles to two-year college squared (÷1,000)	.011 (.002)	-.003 (.002)
Miles to four-year college (÷10)	.016 (.005)	-.026 (.005)
Miles to four-year college squared (÷1,000)	-.010 (.004)	.013 (.004)
Two-year tuition in state (\$82) (÷1,000)	-.093 (.043)	.113 (.044)
Four-year tuition in state (\$82) (÷1,000)	.056 (.033)	-.127 (.034)
<i>F</i> statistic of joint significance of instruments	41.16	14.22
<i>R</i> ²	.105	.271

NOTE: There are 6,786 observations. Also included are a constant, female, race, the HS&B test score, parents' education, family income, whether family owns home, % of high-school enrollment that is White, high school in urban area, high school's city unemployment rate in 1980, census region dummies, and state income per capita in 1980 in state of residence when in high school. The coefficients from the other independent variables are available from the author on request.

Using this strategy I identify the effects of college type on educational attainment by estimating Equation (2). I interpret α_2 , the gain in attainment for those who begin in a community college compared to those with no college, as the effect of democratization on educational attainment. The difference, $\alpha_2 - \alpha_4$, is the effect of diversion because it shows the change in attainment, if there is one, of starting in a junior college instead of a four-year college.

Table 4 presents linear IV estimates of the treatment effects of two-year and four-year colleges. These results are unweighted, although weighting appears to make little differ-

Table 4. The Effect of College Type on Educational Attainment: OLS and IV Results

Model/college type	Dependent variable: years of education	Dependent variable: 1 = bachelor's degree/ 0 = no degree
OLS/linear probability:		
Two-year college (α_2)	.936 (.041)	.059 (.008)
Four-year college (α_4)	2.027 (.044)	.304 (.011)
$\alpha_2 - \alpha_4$	-1.091 (.049)	-.245 (.013)
IV/instruments: miles to two- and four-year college, miles sq.:		
Two-year college (α_2)	1.215 (.422)	.098 (.112)
Four-year college (α_4)	1.859 (.736)	.192 (.192)
$\alpha_2 - \alpha_4$	-.644 (.495)	-.094 (.129)
GMM χ^2 (df)	.124(2)	.434(2)
IV/instruments: miles to two- and four-year college, miles sq./two- and four-year college tuition:		
Two-year college (α_2)	1.488 (.545)	.078 (.103)
Four-year college (α_4)	2.493 (1.025)	.185 (.157)
$\alpha_2 - \alpha_4$	-1.005 (.617)	-.107 (.109)
GMM χ^2 (df)	.635(4)	2.939(4)
IV/instruments: miles to two- and four-year college as 5 dummy variables:		
Two-year college (α_2)	1.530 (.379)	.168 (.098)
Four-year college (α_4)	2.114 (.645)	.248 (.168)
$\alpha_2 - \alpha_4$	-.584 (.437)	-.080 (.114)
GMM χ^2 (df)	14.16(8)	9.901(8)
IV/instruments: miles to two- and four-year college as 5 dummy variables/two- and four-year tuition:		
Two-year college (α_2)	1.699 (.359)	.136 (.092)
Four-year college (α_4)	2.549 (.556)	.210 (.145)
$\alpha_2 - \alpha_4$	-.850 (.387)	-.074 (.101)
GMM χ^2 (df)	15.68(10)	12.59(10)

NOTE: White (1980) standard errors are in parentheses. For other independent variables see Table 3. The critical values (at the .05 level) are 5.99 for two df, 9.49 for four df, 15.51 for eight df, and 18.31 for ten df.

ence. OLS estimates of the effects of college type on attainment that have not been corrected for self-selection are shown in the top row. Using both definitions of attainment, these results suggest that the educational attainment of those who begin in a two-year college is significantly lower than the attainment of those who begin in a four-year college. In other words, the results indicate a negative, and significant, effect of diversion on educational attainment, the results found by others. The IV results occupy the bottom four rows of the table.

The treatment effects are positive and significant in all of the equations explaining years of education. On the other hand, the point estimates in the degree equation, though positive, have large standard errors such that the values could be interpreted as 0. Slightly smaller standard errors in all equations suggest that distance performs better as an instrument once it is allowed a very nonlinear functional form.

The magnitudes of the coefficients on starting in a two-year college suggest a positive effect of democratization on years of schooling that is statistically significant. Those who start in community college receive one to one-and-one-half more years of education than those who do not attend college. The effect of starting in a junior college (the effect of democratization) in the degree equation appears quite small, although the coefficients are much less precisely estimated than those estimating years of education.

The effect of diversion is identified by subtracting the four-year treatment effect from the two-year treatment effect. The coefficients suggest a negative effect of diversion on years of education that is marginally significant when tuition is also included as an instrumental variable. Focusing on the largest coefficient, starting in a two-year college, instead of a four-year college, leads to at most one less year of education. In the degree equation, the effect of diversion is negative but not significant, as a result of large standard errors on both coefficients. Taken at face value, a *T* test indicates that there is no change in the probability of attaining a bachelor's degree from starting at a junior college; those who start in a junior college probably would not have finished a degree anyway.

The test statistics from the generalized method of moments (GMM) error-orthogonality test are reported for each equation. The test involves regressing the residuals from the second-stage equation on all of the exogenous variables. The R^2 of this regression times the number of observations is asymptotically distributed as a χ^2 with $k - q$ df, where k is the number of instruments and q the number of endogenous variables (Newey 1985). The weaker the association between the instruments and the second-stage residuals, the less likely one is to reject the null hypothesis that the equations are properly specified. All of the specifications pass the tests of overidentifying restrictions.

These results call into question some of the earlier findings on the effect of diversion on educational attainment. As found by others, students who were diverted from four-year college appear to lose between one-half to one year of education, although this effect is only statistically significant in one out of four cases. Furthermore it is not clear that they

would have completed a bachelor's degree anyway. It is important to remember that the outcome variable I consider here is total years of schooling. Kane and Rouse (1993) examined the effects of community-college attendance on earnings and found that two-year and four-year college credits result in roughly similar earnings gains relative to high-school graduates with no college education. Thus the treatment effects on years of schooling translate directly into individual and social benefits of two-year and four-year college.

The overall effect of community colleges on educational attainment can be assessed by comparing the magnitudes of the college-type effects weighted by the proportions of students in each group. Rouse (1994) estimated that community-college students are approximately equally divided into each group. Because the magnitude of the effect of democratization is larger than that of diversion, the estimates suggest that the increase in education of providing the two-year college as a port-of-entry to higher education marginally outweighs the decrease in attainment due to diversion. However, *T* tests (not presented here) also indicate that the difference between the two effects is only statistically significant (at the 10% level) in the two cases when distance is divided into five groups. These results are consistent with the overall effects estimated using the reduced-form and are partially consistent with Grubb (1989), who, using state-level data, concluded that community colleges appear to increase the educational attainment of men but not women.

5. ASSESSING THE EXOGENEITY OF COLLEGE ACCESSIBILITY

In Table 4, tests of the overidentifying restrictions were presented to help assess the consistency of the instruments. The test is a method of assessing the validity of an additional instrument assuming that one already has at least one valid instrument. In this instance, however, neither instrument can plausibly be assumed to be truly exogenous. In this section, I further explore the extent to which college proximity and tuition may be correlated with unobservable determinants of educational attainment as well as the resulting bias of the IV estimates.

First, community colleges may be built in areas in which individuals desire one. Cohen and Brawer (1982) cited a study that examined the expansion of community colleges and argued that the institutions were built in areas with high population density. The author found that when 90–95% of a state's population lived within commuting distance (about 25 miles), a community-college system is considered "mature" and few additional colleges are built. If population density grows in a manner that is exogenous to an unobserved desire for education, then distance to the nearest community college is likely to be a valid instrument.

Nevertheless, it is still possible that community colleges were built in lower-income neighborhoods where individuals wanted a community college to complete one to two years of college. To assess the possible bias remaining in the IV estimates due to the correlation between the instruments and

Table 5. IV Results Sequentially Removing Family Background Measures (standard errors)

College type	Base IV estimate	Without father's education	Without father's and mother's education	Without both parents' education and income	Without parents' education, income, and measured ability
Two-year college	1.26 (.67)	1.62 (.48)	1.66 (.42)	1.96 (.41)	2.09 (.38)
Four-year college	2.12 (1.12)	2.72 (.64)	2.83 (.55)	3.09 (.51)	3.28 (.37)

NOTE: IV estimates use distance to two-year and four-year college and two-year and four-year college tuition as instruments. See notes to Table 2 for other covariates. Dependent variable = years of education.

unobserved factors in educational attainment (such as desired level of educational attainment), I assume that an estimate of the correlation between the instruments and the unobserved factors in educational attainment can be obtained from the correlation between the instruments and observed determinants of educational attainment—namely, family background. More specifically, suppose that the true model of educational attainment is represented in Equations (2)–(4) plus the following equation: $S_i = X_i\pi_0 + \pi_2 D_{2i} + \pi_4 D_{4i} + \nu_i$, such that the unobserved desired level of educational attainment is a function of observed family background and measured ability, distance to the nearest two-year and four-year colleges, and a random-error term. Solving for the IV estimate of α_2 leads to $\hat{\alpha}_{2(IV)} = \alpha_2 + \lambda[(\gamma_2 + \lambda_2\pi_2)(D_2'M_4D_2) + (\delta_2 + \lambda_2\pi_4)(D_2'M_4D_4)]^{-1}[\pi_2(D_2'M_4D_2) + \pi_4(D_2'M_4D_4)]$, where $\hat{\alpha}_{2(IV)}$ is the IV estimate of $\hat{\alpha}_2$, $M_4 = [I - A_4(D_4'A_4)^{-1}D_4']$, and a subscript, i , has been suppressed for simplicity. (There is an analogous expression for $\hat{\alpha}_{4(IV)}$.) As π_2 and π_4 increase in magnitude, so does the bias in the IV estimate. Thus one way to gauge the extent of the bias is to assume that observed measures of family background also reflect unobserved determinants of educational attainment. In this case, I can observe changes in $\hat{\alpha}_{2(IV)}$ and $\hat{\alpha}_{4(IV)}$ as various measures of family background and measured ability are removed from the equations. $\hat{\alpha}_{2(IV)}$ and $\hat{\alpha}_{4(IV)}$ will increase as these measures are removed if the observed measures of family background and measured ability are positively correlated with college accessibility. The results are presented in Table 5.

As measures of family background and ability are eliminated, the coefficients on two-year and four-year college increase. To the extent that observed measures of family background also reflect unobserved determinants of educational attainment, there is evidence that the instruments may be correlated with the error term in the educational attainment equation. The ratio of the effect of two-year to four-year college on educational attainment is roughly constant, however. Thus the bias is likely more important for the treatment effect of college (in general) than for the relative effects of college type on educational attainment.

6. CONCLUSION

Distance can serve as an instrument and policy variable in a variety of applications. Nonetheless this research highlights two potential limitations. First, this article cautions

that location decisions are potentially endogenous. In the data set used here, the location decisions were made by the respondent's parents, not the respondents, which increases the validity of the instruments. Nonetheless, the case for exogeneity can only be made conditional on family background and location effects. Furthermore, the relatively large standard errors of the IV estimates presented suggest the need for much data to isolate treatment effects with precision. For example, due to the large standard errors, one could not reject the hypothesis that the IV estimates are statistically similar to the OLS estimates.

Previous studies have cast doubt on the benefits of community colleges in higher education. Critics assert that this second track of college diverts able students who would complete more schooling in a four-year institution. Advocates argue that community colleges are low-cost substitutes for the first two years of a four-year education. As states consider expanding their community-college systems as an antidote to the increasing cost of providing public higher education, it is essential that we better understand the costs and benefits of these institutions.

This article updates the results of previous studies and derives the first estimates of the effect of college type on educational attainment from a strategy that attempts to isolate a consistent treatment effect. Using city and state variation in proximity to community colleges, I find that a closer local community college will increase total years of schooling slightly and not likely change the likelihood of receiving a bachelor's degree. This occurs because the community college will divert some students from four-year schools who would have completed more years of schooling in a four-year college; this effect is marginally outweighed by the increase in educational attainment by those who otherwise would not have attended college at all. I also find that those diverted from four-year college probably would not have finished a bachelor's degree anyway. Given that current fund expenditures per full-time equivalent student in public four-year institutions is generally twice that of public two-year schools, the results here suggest that community colleges have a potential to increase overall educational attainment that deserves greater consideration (*Digest of Education Statistics* 1990, p. 301).

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APPENDIX: DATA

High School and Beyond (HS&B) is a survey conducted by the Department of Education to complement the National Longitudinal Survey of the High School Class of 1972. Employing a two-stage sampling design in which high schools were randomly chosen in the first stage and in which students were randomly sampled from within each high school in the second, HS&B included about 12,000 seniors from the high-school class of 1980 from about 1,100 schools. Schools with relatively more minority and low-income students were over-sampled. Follow-up surveys were conducted in February 1982, 1984, and 1986, and by 1986 the survey retains 88% of the original base sample.

To calculate completed years of education, I started by assigning 12 years of education to all members of the survey because the survey was conducted during the spring of their senior year in high school. Each additional year in a post-secondary institution counted as one year. I assigned a vocational degree 13 years, an associate degree 14 years, and a bachelor's degree 16 years (even if it took the respondent five years to complete). Those who have enrolled in a graduate program were given 17 years, and a graduate degree corresponded to 18 years of education.

The tuition data for both two-year and four-year colleges are from the State of Washington Higher Education Coordinating Board.

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